

Surface Area of Pyramids

Lesson 10-5

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

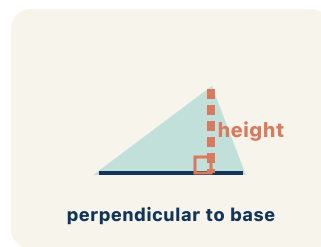
Picture first, then the word, then a plain-language meaning. Say each word out loud.



Like the Great Pyramid of Giza — a square on the bottom, 4 triangles slanting up to a point

Pyramid

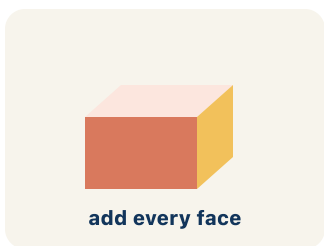
Write the definition:



If you slide your finger from the bottom edge up the triangle face to the top point — that distance is the slant height

Slant height

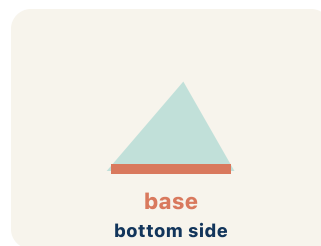
Write the definition:



A square pyramid has 4 lateral faces — one triangle for each side of the square base

Lateral face

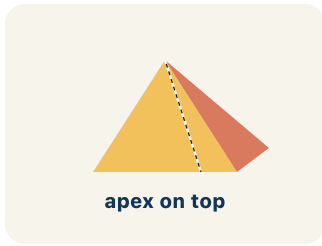
Write the definition:



A square pyramid sits on a square base; base area = side × side

Base

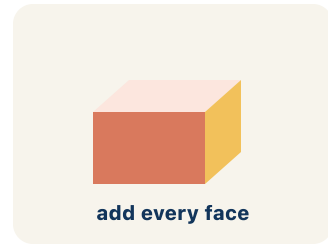
Write the definition:



The pointy top of a pyramid — all the slanted edges connect here

Apex

Write the definition:



For a square pyramid: lateral area = $4 \times (\frac{1}{2} \times \text{base edge} \times \text{slant height})$

Lateral area

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find the surface area of a pyramid by adding the base area and the lateral faces.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Pyramid

Slant height

Lateral face

Base

Apex

Lateral area

1 A solid with a polygon base and triangular faces meeting at one point is a _____.

2 The height of a triangular face of a pyramid, measured along the slanted side, is the _____.

3 A triangular side face of a pyramid is a _____.

4 The polygon on the bottom of a pyramid is the _____.

5 The single top point where the triangular faces of a pyramid meet is the _____.

6 The combined area of all the side faces, not counting the base, is the _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: A square pyramid has a base edge of 5 in and a slant height of 7 in. What is the surface area?

- A. 95 in^2
- B. 70 in^2
- C. 95 in^3
- D. 120 in^2

Step 1 Base: $5 \times 5 = 25 \text{ in}^2$.

Step 2 Each lateral face: $\frac{1}{2} \times 5 \times 7 = 17.5 \text{ in}^2$.

Step 3 Four faces: $4 \times 17.5 = 70 \text{ in}^2$.

Step 4 $SA = 25 + 70 = 95 \text{ in}^2$.

 **Answer:** A. 95 in^2

 We do — together

Solve this one with your class using the same steps.

Problem: How many lateral (triangular) faces does a square pyramid have?

- A. 4
- B. 3
- C. 5
- D. 6

Step 1 _____

Step 2 _____

Step 3 _____

Answer: _____

**You do — your turn**

Now try one on your own. Show every step.

Problem: What is the area of one triangular face with base 8 in and slant height 6 in?

A. 24 in^2

B. 48 in^2

C. 14 in^2

D. 24 in^3

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Before adding the base, you need the area of just one triangular face of a gift box with base edge 8 in and slant height 5 in. What is it?

- A. 20 in^2
- B. 40 in^2
- C. 13 in^2
- D. 20 in^3

Show your work:

2. A gift box is a square pyramid with a base edge of 9 in. What is the area of its square base (the part that sits on the table)?

- A. 81 in^2
- B. 36 in^2
- C. 18 in^2
- D. 324 in^2

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two pyramids have the same base area (64 in^2) but different slant heights. Pyramid A has slant height 6 in and Pyramid B has slant height 10 in. How much more surface area does Pyramid B have? Why does slant height affect surface area but NOT base area?

Sentence starter: Pyramid A: $SA = 64 + 4(\frac{1}{2} \times 8 \times 6) = 64 + \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ in}^2$. Pyramid B: $SA = 64 + 4(\frac{1}{2} \times 8 \times 10) = 64 + \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ in}^2$. Pyramid B has $\underline{\hspace{1cm}}$ more in^2 because $\underline{\hspace{1cm}}$. Slant height only affects $\underline{\hspace{1cm}}$ because $\underline{\hspace{1cm}}$.

Show your work:

Reflect — Exit Ticket

A square pyramid has a base edge of 6 in and a slant height of 5 in. What is the total surface area?

- A. 96 in^2
- B. 60 in^2
- C. 96 in^3
- D. 36 in^2

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Pyramid
2. **Notes 2:** Slant height
3. **Notes 3:** Lateral face
4. **Notes 4:** Base
5. **Notes 5:** Apex
6. **Notes 6:** Lateral area
7. **We Try (notes):** A. 4 — *A square pyramid has a square base and 4 triangular lateral faces — one for each side of the square.*
8. **You Try (notes):** A. 24 in^2 — *Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 6 = 24 \text{ in}^2$.*
9. **Try It 1:** A. 20 in^2 — *Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 5 = 20 \text{ in}^2$. You use the slant height (5) as the triangle's height, not the pyramid's vertical height.*
10. **Try It 2:** A. 81 in^2 — *The base is a square, so base area = side $\times$ side = $9 \times 9 = 81 \text{ in}^2$. You would still add the four triangular faces to get the full surface area.*
11. **Exit Ticket:** A. 96 in^2 — *Base: $6 \times 6 = 36 \text{ in}^2$. Each lateral face: $\frac{1}{2} \times 6 \times 5 = 15 \text{ in}^2$. Four faces: $4 \times 15 = 60 \text{ in}^2$. $SA = 36 + 60 = 96 \text{ in}^2$. Surface area uses square units (in^2).*